HOUSING SOLUTIONS KFT.

HR-ERP Platform

AI-powered, multilingual workforce & housing management

Hungarian HR. Native-language workers. One platform.

Seed Investment Opportunity · 2026-04-19 · v3

Lerch Balazs — Founder & CEO

The Problem: Language Barrier + Turnover

Hungarian employers run Filipino, Ukrainian and German crews — using paper and translators

30–50%

Annual turnover

In blue-collar crews — €5–15K per exit in recruitment, training and lost productivity

60%

HR time on "firefighting"

Housing complaints, cross-language chat, damage claims, paper admin

5

distinct languages

on one site: Hungarian, English, Tagalog, Ukrainian, German — misunderstanding is the default

The gap: global HR platforms (Workday, ADP) don't know Hungarian labour law, don't run worker housing, and can't hold a conversation with a Filipino worker. Local point tools don't integrate.

€500,000+

annual loss per 100-person site
paper processes, miscommunication, lost documents

Our Solution: Integrated HR-ERP + AI

One platform from ticket intake to payroll — with native-language communication

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Housing + HR together

Room assignments, bed occupancy, damage reports — tied to the employee record

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Tickets + SLA

Auto-assignment, SLA deadlines, escalation — mobile-first submission

🤖

AI multilingual layer

Claude Haiku 4.5 — the Filipino worker writes Tagalog, the HR lead reads Hungarian

💙

WellMind + CarePath

Daily pulse surveys, NLP sentiment, 24/7 EAP advisory

📈

Finance + OCR

Claude OCR invoice processing, cost-center tree, salary band management

The only Hungarian HR platform that unifies worker housing, the HR record and AI-powered native-language communication.

Platform Scale — Not an MVP, Production

Measured as of 2026-04-19 — 27 commits in the last 30 days

17 functional modules	**427** API endpoints	**124** DB tables (live)	**423** DB indexes (live)
74 SQL migrations	**56** test files	**64** admin UI pages	**51** mobile screens

Code footprint

- **169** backend JS files (Node / Express)
- **~1,600** i18n keys (319 × 5 languages)
- **+28** new API endpoints, **+21** new tables in 30 days
- Test count nearly doubled: 31 → **56** files

Production maturity

- Kubernetes deployment + GitHub Actions CI/CD
- Redis cache + Bull job queue for async work
- PWA admin UI, ETag HTTP caching
- Daily log rotation, 30-day retention
- 1 SQL injection fixed (audit baseline)

🤖 AI & Multilingual — The Core Differentiator

Claude Haiku 4.5 powered real-time translation across 5 languages

5 supported languages

🇭🇺	🇬🇧	🇵🇭	🇺🇦	🇩🇪
Hungarian	**English**	**Tagalog** Filipino	**Ukrainian**	**German**

How it works

- The worker writes the ticket / comment in their **native language**
- The system auto-translates to the viewer's `preferred_language`
- The Hungarian admin reads everything in Hungarian — no Google Translate copy-paste
- Two-way: admin replies also appear in the worker's language

Engineering detail

- **Model:** `claude-haiku-4-5-20251001`
- **Prompt caching** on the system prompt — ~90% input-cost reduction
- **30-day DB cache** (`translation_cache` table) — repeated strings are free
- **XML-tagged input** — prompt-injection protection
- Fallback: Google Translate (if Claude API unreachable)

Cost profile

~90%
cost reduction from prompt caching + DB cache combined

2×
Claude integration: translation + invoice OCR

Wired in: ticket viewer, chatbot, damage report (migration written), mobile push notifications

🦄 Bruno AI — Context-Aware Onboarding Companion

An animated AI character that coaches users mid-workflow — especially new foreign workers

What Bruno does

- **Context-aware tips:** sees the current page and offers targeted advice
- **Animated reactions:** motivating feedback paired with the gamification engine
- **Speaks the viewer's language:** follows each user's language preference
- **Non-intrusive:** only appears when useful — new user, stuck flow, new feature

Why it matters

A Filipino or Ukrainian worker rarely gets a proper onboarding on day one. Bruno steps in: shows how to file a ticket, where to complete the pulse survey, what to do when a damage report lands on their screen.

Under the hood

- `components/Bruno/BrunoCharacter.jsx`
- 296 lines of CSS — full animation library
- Admin entry point: `pages/BrunoTest.jsx`
- Drop-in on every admin or mobile screen
- Claude-powered answers when the user asks a question

Result: shorter onboarding, fewer HR tickets, higher adoption — especially for new workers with a language barrier.

17 Modules — One Platform

Seven functional groups, all wired in and running in production

Core

- Tickets (SLA, auto-assignment)
- Employee records (PII-encrypted)
- Accommodations + room assignment
- Contractor management

Finance

- Payroll + salary band management
- Invoicing + PDF export
- Invoice OCR (Claude, Gmail polling)
- Cost-center tree

Wellbeing

- WellMind — pulse survey (86 questions)
- Pulse Analytics Dashboard (5 views)
- CarePath — 24/7 EAP
- NLP sentiment analysis (opt-in)

Productivity

- GTD Task Manager (new)
- Projects + Kanban + Gantt
- Timesheet across 3 views
- SLA deadline tracker

Communication

- AI Chatbot (Claude)
- Multilingual translation (5 languages)
- Notification Center
- Slack integration

Compliance

- Damage reports (Mt. § 166, § 177)
- Full audit log
- GDPR data-controller role
- 5-language PDF generation

Engagement: Gamification (badges, leaderboard, streaks), Bruno AI context-aware character, camera-based document scanning, Google Calendar sync.

Technical Architecture

Production-ready, not a prototype

Stack

Layer	Technology
Admin UI	React 18 + Vite 5 + MUI 5 (PWA)
Mobile	React Native 0.81 + Expo 54
Backend	Node.js + Express 4.18 (cluster mode)
DB	PostgreSQL, **124 tables**, **423 indexes**
Cache / Queue	Redis (ioredis) + Bull 4.16
AI	Anthropic Claude (translation + OCR + chatbot)
Deployment	Kubernetes + GitHub Actions CI/CD
i18n	react-i18next, 5 languages, ~1,600 keys

Integrations

- **Claude API** (two touch-points): `translation.service.js` + `claudeOCR.service.js`
- **Google Calendar** two-way sync + webhook
- **Gmail MCP** — invoice email polling
- **Slack** (@slack/bolt) — alerts, user sync
- **SMTP** (nodemailer) — transactional email
- **Tesseract.js** OCR fallback

Performance

PWA
offline-first admin UI

ETag
HTTP cache on every GET

Bull
async background queue

RLS
multi-tenant isolation

Security & Compliance

GDPR + EU AI Act compliant — engineering-grade, not marketing-grade

Security stack

- **JWT** auth, strong secret (rotated post-audit)
- **CSRF** double-submit cookie protection
- **Rate limiting** + slow-down (login: 5 / 15 min)
- **Helmet** + HSTS + CSP (`/api/v1/csp-report`)
- **Field-level PII AES encryption** (tax ID, address, SSN)
- **Row-Level Security** for multi-tenant isolation
- **bcrypt** + password policy

Fixed in the last 30 days

1 SQL injection fixed — `setDatabaseUser.js` template-literal queries converted to parameterized, commit `2e943991`. **Strong JWT secret** generation replacing the weak default.

Compliance

Regulation	Status
GDPR Art. 22 / 25 / 35	✔ Compliant
EU AI Act	✔ Compliant
Mt. § 166 (damage liability, HU)	✔
Mt. § 177 (damage claim, HU)	✔
ISO 45003	✔
DPIA documented	✔

Audit and roles

- **7 roles:** superadmin, data_controller, admin, task_owner, contractor, user, accommodated_employee
- **40+ base permissions** (`module.action` format)
- **Full audit trail** on every write
- **Key rotation** for PII encryption (`rotate-encryption-key.js`)

🚀 Competitive Advantage

"The only Hungarian HR platform that..."

✔ **Runs on native-language AI translation** — Claude Haiku 4.5, 5 languages, 30-day cache, prompt caching. The Filipino worker writes Tagalog; the Hungarian HR lead reads Hungarian.

✔ **Treats housing AND the workforce as one system** — room assignment, bed occupancy, damage reports, cleanliness inspections, all tied to the employee record. No parallel spreadsheet.

✔ **Provides end-to-end wellbeing monitoring** — WellMind pulse (86 questions) + Pulse Analytics Dashboard + CarePath EAP + opt-in NLP sentiment. Prevention, not reaction.

✔ **Is GDPR + EU AI Act compliant by design** — dedicated data-controller role, opt-in sentiment, DPIA on file, Mt. § 177 damage-claim workflow with 5-language PDF.

✔ **Lives on production Kubernetes infrastructure** — not an MVP, already running. 124 tables, 423 indexes, 427 API endpoints, 56 test files, CI/CD pipeline. 27 commits in 30 days.

Market & ROI

Hungarian and CEE market — underserved, with strong customer-side ROI

Target market

€500M+

CEE SMB + mid-market employers running blue-collar crews with employer-provided housing

ICP (Ideal Customer Profile)

- Construction, hospitality, manufacturing, logistics
- 50–5,000 employees, meaningful share foreign
- Employer-provided housing on site
- High turnover (>30%)
- Hungary → Poland, Czechia, Slovakia

Pricing (SaaS, modular)

Plan	Per person / month
Core (housing + tickets + mobile)	€3
+ WellMind	+€1
+ CarePath	+€1

Customer ROI — example (200 workers)

Saving	Annual
Turnover reduction (40% → 25%)	**€900,000**
Sick-leave reduction (20%)	**€30,000**
Productivity gain (10%)	**€480,000**
Total	**€1,410,000**
Platform cost	€12,000
ROI	**117×**

Unit economics

- Gross margin: **85%**
- Target LTV : CAC: **9–15 : 1**
- Payback: **< 2 months**
- Target annual churn: **< 10%**

Roadmap — Next 6–12 Months

The platform is live. Next steps: GTM and deepening the translation layer

Q2 2026

Pilot + i18n depth

- Mobile language picker UI
- Damage-report translation wired in
- 2–3 pilot customers, case study
- Ship migration 076 to prod

Q3 2026

PMF + multilingual chatbot

- Chatbot replies in viewer's language
- Scheduled reports (node-cron)
- 10 customers, €15–30K MRR
- Pricing validation

Q4 2026

Scale

- 50 customers, €60–120K ARR
- CEE expansion groundwork
- Hire 1–2 sales reps
- Kick off SOC 2 Type II audit

2027

CEE leader

- Poland, Czechia, Slovakia
- €165K+ ARR
- Extra languages: PL, CS, SK
- Category-defining position

Investment ask

Seed Round

€500,000

Pre-money valuation: €2.5M

- **Sales + Marketing — 40% (€200K)**
- **Product development — 30% (€150K)**
- **Operations + Legal — 30% (€150K)**

Exit paths

- **Strategic acquisition:** Workday, SAP SuccessFactors, ADP — CEE beachhead
- **Regional HR-SaaS consolidator** (BambooHR, Personio)
- **Private Equity:** 8–12× ARR (HR-vertical roll-up)

Runway

12 months to product-market fit (10 paying customers, <10% churn).

Break-even MRR: ~€6,500 / month, expected month 8–9.

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Production-ready AI HR-ERP. Raising seed now.

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17	427	124	5	K8s
Modules	API endpoints	DB tables	Languages	Production

Claude Haiku 4.5 · Redis + Bull · PWA · GDPR + AI Act · 27 commits / 30 days